

BabyHuBERT: Multilingual Self-Supervised Learning for Segmenting Speakers in Child-Centered Long-Form Recordings

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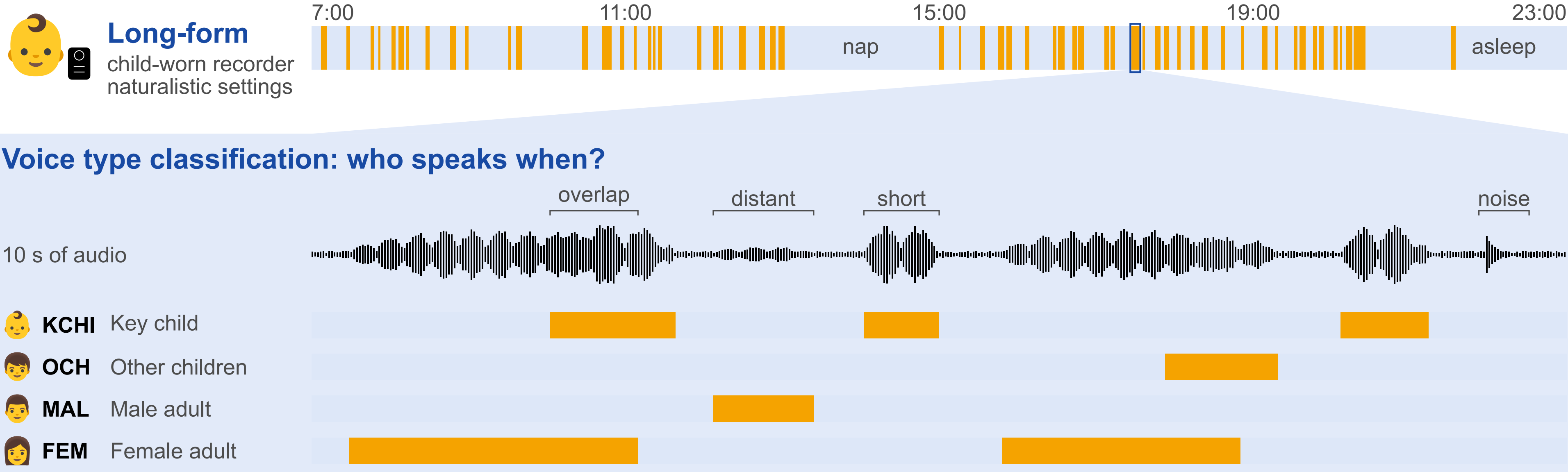
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Context and problem statement

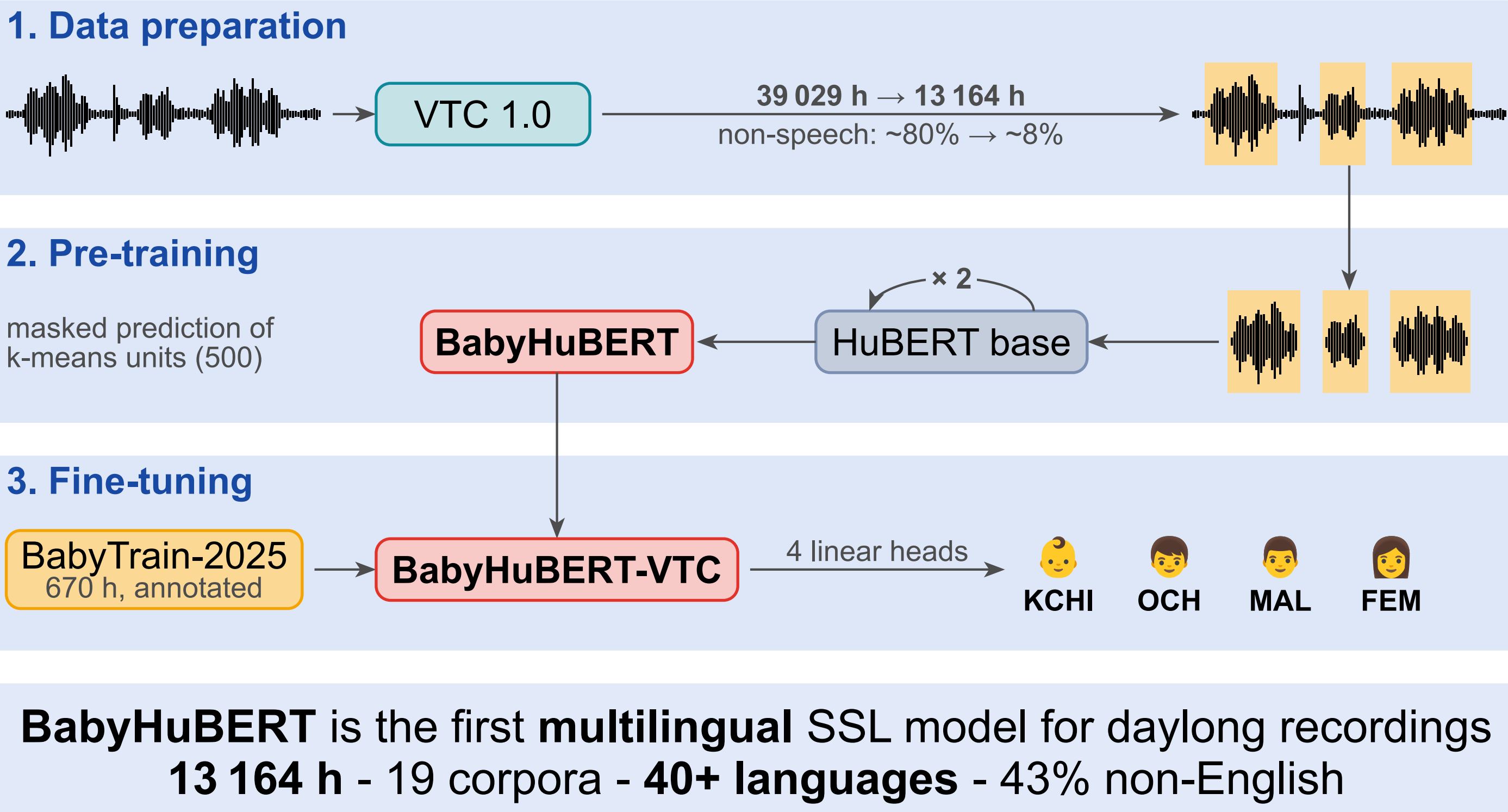
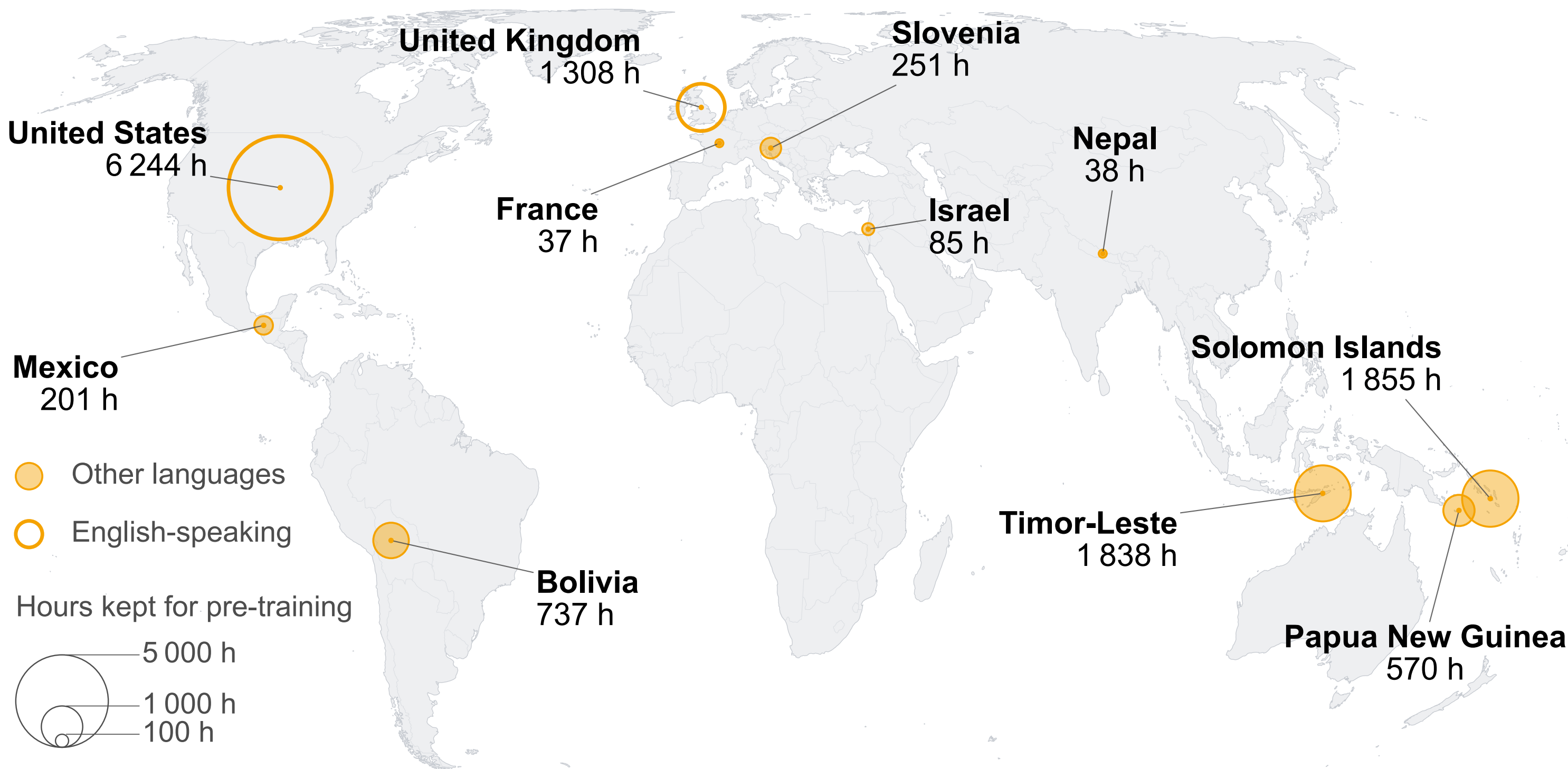
Child-centered **daylong recordings** are our best window onto the speech children hear and produce.

But analyzing 16-hour recordings of noisy, naturalistic child-caregiver interactions **by hand does not scale**.

Existing approaches still largely rely on SSL models pre-trained on **monolingual clean adult speech** (domain mismatch).

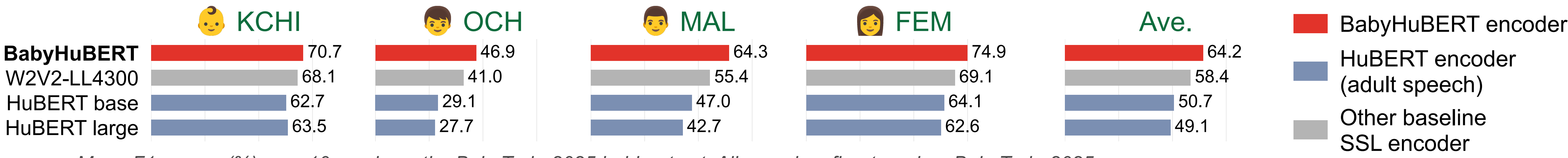


Data and training



Result 1 How do different SSL models compare?

BabyHuBERT largely outperforms out-of-domain (HuBERT) and English-only in-domain (W2V2-LL4300) SSL models

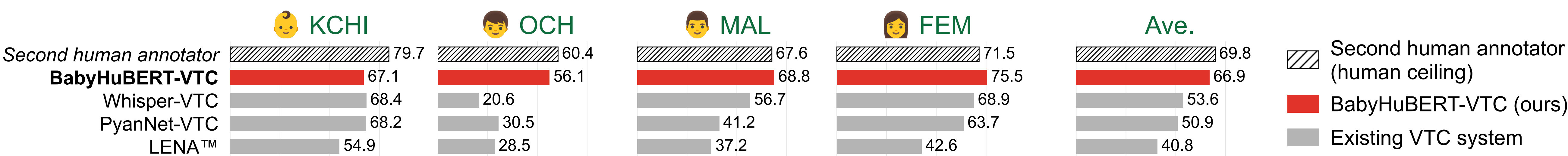


Result 2 How does it perform compared to the state of the art?

BabyHuBERT-VTC outperforms all existing VTC systems (+13.3pp F1 over Whisper-VTC)

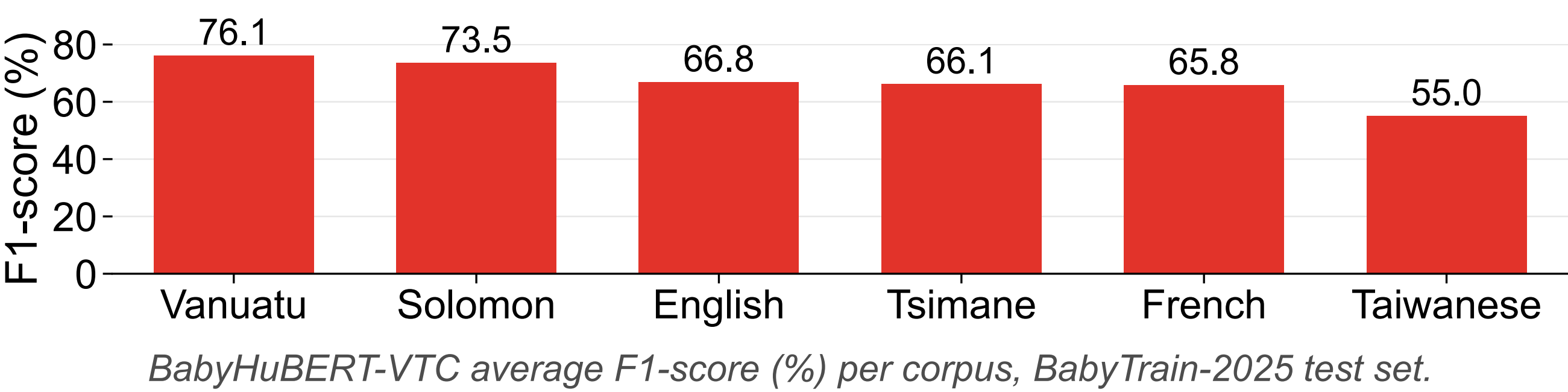
OCH F1 nearly doubled: 30.5% → 56.1%

Human-level on MAL & FEM, close on OCH (vs. a 2nd annotator)



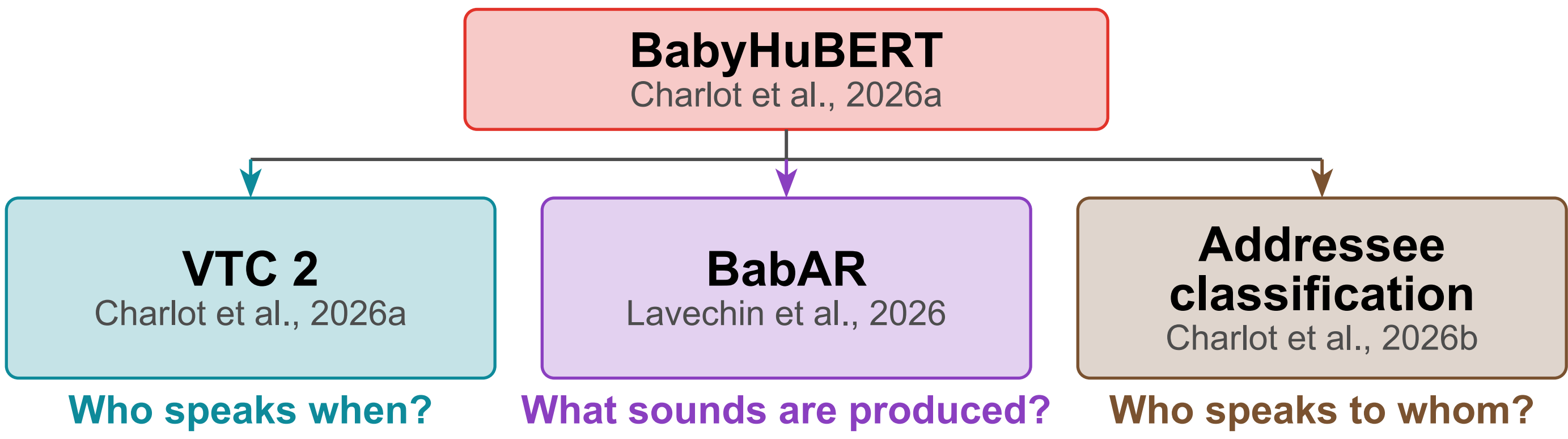
Result 3 How does it perform across languages?

Majority languages do not necessarily perform best
Highest F1 on the multilingual Vanuatu (76.1%) and Solomon Islands (73.5%) corpora;
English is 57% of the pre-training data and only ranks third (66.8%).



Outlook BabyHuBERT's downstream applications

BabyHuBERT as a foundation for developmental speech research



References

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